

Closed-Loop Streaming Synthesis Bounded-Drift, Arbitrary-Length Audio–Video Generation by Treating Chunk Hand-off as Feedback Control

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Abstract

Generating arbitrarily long videos with diffusion transformers requires splitting the temporal dimension into overlapping chunks. Naive open-loop chunk-streaming accumulates exposure-bias drift: each chunk is conditioned on the previously generated latent, compounding distributional errors and causing scene collapse or grain amplification after a few hundred frames. We present **Closed-Loop Streaming Synthesis (CLSS)**, a suite of four post-denoising corrections that close the feedback loop between successive chunks without modifying the transformer weights: calibrated context re-noising, per-channel EMA-tracked AdaIN, frequency-band soft shrinkage, and a dynamic anchor bank. CLSS is implemented as a ComfyUI custom-node pipeline targeting LTX-2.3 22B (a 44 GB audio-video diffusion transformer) on consumer 16 GB VRAM hardware via GGUF 4-bit quantisation and CPU block-streaming. Generation is driven by a reference-parity multi-modal guider (split per-modality classifier-free guidance, modality-isolation guidance, and spatiotemporal guidance via a value-passthrough self-attention skip) that correctly handles ComfyUI’s packed audio-video latent representation, and all streaming noise is seeded for bit-reproducible runs. A two-stage pipeline generates a draft audio-video via CLSS streaming and then refines it using a 3-step distilled-LoRA schedule at $2\times$ spatial resolution.

We report matched text-to-video (T2V) and image-to-video (I2V) experiments: single-scene, ≈ 53 -second clips generated as ten chunks ($\text{new_lf} = 16$, $\text{overlap} = 8$) under an identical seed and guidance stack (split per-modality CFG 4.0 video / 7.0 audio, STG 1.0 on both modalities at block 28, rescale 0.70; three transformer passes per step), differing only in the presence of a guide image. Stage 1 holds video boundary cosine similarity at 0.84–0.98 in I2V; a two-band spatial detail anchor holds the high-frequency energy share flat across the full run in both modes ($0.498 \rightarrow 0.498$ I2V, $0.512 \rightarrow 0.520$ T2V), closing the

progressive-softening channel, and a reference-exact RMS anchor holds audio loudness to its chunk-1 level throughout ($\pm 5\%$) against raw per-chunk drifts of up to $+72\%$ (T2V). Stage 2 ($2\times$ resolution, 3-step distilled refinement in four windows) refines video against frozen Stage-1 audio, with its own per-window detail anchor referenced to the upscaled Stage-1 slice; end-of-window fidelity to Stage-1 structure stays at 0.86–0.94 (I2V). The principal audio failure mode — a per-chunk loudness-envelope repetition — is eliminated: its carrier is the near-verbatim audio overlap tail, and re-noising the audio SLB on a hotter schedule with an envelope-flattened context keeps the envelope correlation below the repetition flag at every seam while preserving seam continuity; audio high-frequency energy growth over the run is characterised separately. Total generation time is ≈ 116 minutes per clip in either mode on a single 16 GB GPU.

1 Introduction

Modern video diffusion transformers operate on patchified latent representations where temporal resolution is down-scaled $8\times$ relative to pixel frames. Fitting a 30-second clip (81,840 tokens at 22×40 spatial) into a single forward pass requires very large GPUs. The standard remedy is *temporal chunking* via a Streaming Latent Buffer (SLB): generate the video in short overlapping segments, conditioning each on the previous denoised output. Section 2 reviews this mechanism and the exposure-bias problem it introduces.

CLSS closes the feedback loop between chunks by applying five lightweight post-denoising corrections that together bound the closed-loop gain below one, preventing drift from accumulating without modifying the transformer weights. The corrections are applied purely at the boundary between chunks and add negligible computational overhead.

Contributions.

1. Five closed-loop correction modules for temporal chunk streaming: calibrated context re-noising, EMA-tracked per-channel AdaIN, frequency-band soft shrinkage, a dynamic anchor bank, and a two-band spatial detail anchor that closes the progressive-softening channel the one-sided shrinkage leaves open (Sections 3.1–3.5).
2. A linear stability analysis of the feedback loop in latent space, including an empirical per-chunk gain measurement protocol (Section 3.6).
3. A memory-efficient two-stage pipeline for LTX-2.3 22B on 16 GB VRAM using GGUF 4-bit quantisation and CPU block-streaming (Section 5).
4. A reference-parity multi-modal guider for ComfyUI (split per-modality CFG, modality-isolation guidance, and STG via value-passthrough self-attention skip) that correctly handles ComfyUI’s packed audio-video latent representation (Section 5.5).
5. ComfyUI custom nodes implementing the full pipeline with seeded, bit-reproducible streaming noise and per-chunk reproducibility telemetry: unconditional settings

dumps, per-frame origin/layout drift event tables, and a phase-lock repetition detector (Sections 5.4, 7.6).

6. Matched ≈ 53 -second T2V and I2V generations, identical seed and guidance stack, streamed as ten chunks each, isolating the effect of visual conditioning on continuity and repetition, and characterising the residual failure modes at this horizon — including a loudness-envelope repetition shown by intervention to be structural to chunked autoregressive audio rather than tunable by any conditioning or guidance knob (Sections 6, 7.6).

2 Background

2.1 LTX-2.3 Architecture

LTX-2.3 is a 22-billion-parameter diffusion transformer for joint audio-video synthesis. It operates on a latent space with an $8\times$ temporal downscale (video VAE, causal) and a $160\times 4\times$ audio downscale (audio VAE, 16 kHz, hop length 160, compression factor 4), yielding a video latent rate of $25/8 = 3.125$ lf/s and an audio latent rate of $16000/(160\times 4) = 25.0$ lf/s. The model accepts interleaved video and audio tokens together with Gemma-3 text embeddings and is trained with flow-matching on the rectified-flow schedule.

2.2 Streaming Latent Buffer (SLB)

The SLB provides temporal continuity in chunk streaming. When generating chunk N , the last `overlap_lf` latent frames from chunk $N - 1$ are prepended to the new chunk’s latent, and the denoiser is conditioned on them at noise level τ_c :

$$\tilde{L}_{\text{overlap}} = (1 - \tau_c) \hat{L}_{\text{overlap}} + \tau_c \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I) \quad (1)$$

Setting $\tau_c = 0$ yields maximal continuity but maximal drift risk; $\tau_c = 1$ decouples the chunks entirely. In LTX’s framework this is implemented as `VideoConditionByLatentIndex` with `strength = 1 - \tau_c`.

Implementation note: masked timestep, not a one-shot draw. Eq. (1) describes the target distribution of the overlap context, but the sampler does not literally draw ε once and hand the model a truncated schedule from τ_c . Tracing the actual mechanism (`ComfyBaseModel.process_timestep` for the LTX model, which the `noise_mask/strength` produced by `VideoConditionByLatentIndex` feeds into) shows that the per-token timestep supplied to the transformer at every sampling step t is $\tau_{\text{eff}} = (1 - \text{strength}) \cdot \sigma_t = \tau_c \sigma_t$ for overlap tokens, versus σ_t for the rest of the sequence — a *persistent* per-step rescaling of the global noise schedule, applied at every step, not a single noise-injection event applied once before an otherwise-standard remaining schedule. For a *linear* (rectified-flow) interpolation path, $x_t = (1 - t)x_0 + t\varepsilon$, these two descriptions coincide exactly: scaling every step’s effective time by a constant factor τ_c is algebraically identical to drawing x_{τ_c} once via Eq. (1) and continuing

the *same* linear ODE from τ_c down to 0. LTX-2.3 is trained with a rectified-flow objective (Section 2), so the equivalence holds for this model; it would not hold in general for a non-linear (e.g. cosine or EDM) noise schedule, where per-step time-rescaling and one-shot truncated-schedule sampling are not interchangeable.

2.3 Exposure Bias in Streaming Diffusion

Let $\hat{L}_N$ be the denoised latent of chunk N and $f_\theta(\cdot \mid \hat{L}_{N-1})$ the denoising operator conditioned on the previous chunk. In open-loop streaming:

$$\hat{L}_N = f_\theta(\hat{L}_{N-1} + \varepsilon_N) \quad (2)$$

The transformer was trained on latents drawn from the data distribution, but at inference it receives denoised latents from prior chunks, which lie off-manifold relative to its training distribution. Any distributional shift $\delta_N = \hat{L}_N - L_N^*$ (where L_N^* is the “on-manifold” latent for the same content) is in general amplified by the open-loop gain of the denoising process. Formally, if $g = \|\partial f_\theta / \partial L\|$ evaluated at L_{N-1}^* , then $\|\delta_{N+1}\| \approx g \|\delta_N\|$. When $g > 1$ this gives $\|\delta_N\| \sim g^N \rightarrow \infty$.

In practice g is not a global constant: it depends on the noise level schedule, step count, guidance strength, and the spectral content of the latent. Section 3.6 describes how g is estimated empirically per chunk via a perturbation measurement.

3 Closed-Loop Streaming Synthesis

CLSS adds five closed-loop correction modules that together bound the error accumulation by ensuring $\rho_{\text{closed}} = g \cdot (1 - \beta) \cdot (1 - \tau_c) < 1$.

3.1 Calibrated Context Re-Noising

The overlap latent fed as context to chunk N is re-noised to level τ_c via Eq. (1). This forces the denoiser to actively re-project the boundary context onto the data manifold rather than accepting it verbatim, partially correcting accumulated drift in the SLB.

In practice $\tau_c = 0.05$ provides a strong temporal constraint (boundary cosine similarity > 0.97) while allowing enough distributional repair to prevent drift accumulation. Larger values ($\tau_c = 0.20$, as suggested by some prior work) loosen the constraint too much for 22B GGUF streaming, causing visible motion discontinuities.

3.2 EMA-Tracked Per-Channel AdaIN

A slow exponential moving average (EMA) with rate $\lambda \approx 0.10$ tracks per-channel mean μ_c and standard deviation σ_c across chunks:

$$\mu_c^{(N)} = (1 - \lambda) \mu_c^{(N-1)} + \lambda \hat{\mu}_c^{(N)} \quad (3)$$

$$\sigma_c^{(N)} = (1 - \lambda) \sigma_c^{(N-1)} + \lambda \hat{\sigma}_c^{(N)} \quad (4)$$

where $\hat{\mu}_c^{(N)}, \hat{\sigma}_c^{(N)}$ are the per-channel statistics of chunk N ’s new frames. After generation, the overlap region is dropped and *only the new frames* L_N^{new} (Algorithm 1, line 5) are blended toward the EMA reference via AdaIN with blend factor β :

$$\tilde{L}_N^{\text{new}} = (1 - \beta) L_N^{\text{new}} + \beta \text{AdaIN}(L_N^{\text{new}} \rightarrow \mu^{(N-1)}, \sigma^{(N-1)}) \quad (5)$$

The overlap region itself is never re-touched by AdaIN — it was already corrected as *previous* chunk’s L_{N-1}^{new} before being pushed to the SLB. This suppresses fast per-chunk statistical drift while allowing slow intentional scene evolution (lighting changes, colour grading) to pass through.

Two safety caps prevent over-correction:

- **Sigma max drift:** the EMA σ_c is clamped to at most $(1 + \Delta) \times \sigma_c^{(0)}$ (default $\Delta = 0.05$), preventing AdaIN from amplifying late-chunk residual noise.
- **AdaIN max amplification:** per-channel upward scaling is capped at $1.2\times$ the current chunk’s std (default), preventing AdaIN from boosting noise in channels that have become quiet.

3.3 Frequency-Band Soft Shrinkage

The latent is decomposed into $B = 4$ frequency bands via 3-D FFT along the (frame, height, width) axes. Each band b receives a continuous gain:

$$g_b^{(c)} = \min\left(1, \left(\frac{E_{\text{ref},b}^{(c)}}{E_b^{(c)}}\right)^{\gamma_b}\right) \in (0, 1] \quad (6)$$

where $E_b^{(c)}$ is the mean squared amplitude of band b in channel c , $E_{\text{ref},b}^{(c)}$ is the corresponding EMA reference, and γ_b is the per-band shrinkage exponent.

Only bands with excess energy relative to their EMA are attenuated. Bands with $\gamma_b = 0$ (the DC band) are never attenuated. Default exponents: $\gamma = (0.0, 0.05, 0.2, 0.3)$, which are conservative enough to preserve scene structure while suppressing temporal flicker.

The EMA is updated with the *corrected* (post-gain) energies, not the raw ones, so the reference tracks intentional scene content rather than accumulated error.

3.4 Dynamic Anchor Bank

A small bank of $M \leq 8$ keyframe anchors provides long-range visual identity conditioning. The bank is seeded from the *last* frame of chunk 1 (not the first) and updated when two consecutive chunks trigger a scene-change signal (cosine similarity below threshold $\rho_a = 0.78$ for $P = 2$ consecutive chunks). Additionally, a *forced insertion* every F_a chunks ($\text{chunk_index} \bmod F_a = 0$, chunk 1 always seeds regardless) prevents bank stagnation on visually stable sequences where the similarity threshold is never triggered. The reference default is $F_a = 5$; the ComfyUI node auto-derives $F_a = \max(2, \min(5, \lceil \text{num_chunks}/4 \rceil))$, giving $F_a = 3$ for the ten-chunk experiments in Section 6.

Anchors are retrieved via cosine similarity and injected as `VideoConditionByKeyframeIndex` at frame index 0 of the current chunk. Anchors too similar to the current overlap ($\text{sim} > 0.85$) are skipped to avoid duplicating the SLB conditioning. LRU eviction discards the least recently used anchor when the bank is full.

3.5 Two-Band Spatial Detail Anchoring

The shrinkage of Section 3.3 is deliberately one-sided (Eq. (6) only ever attenuates), and this asymmetry has a measured long-run consequence: low-band energy inflates while high bands are only ever shrunk, so the *high-frequency share* of latent energy falls monotonically — +19.6% band-0 inflation over seven chunks in one measured run, with the uncapped band EMA itself drifting upward and legitimising the excess. Nothing in the shrinkage loop can *restore* band energy, and the perceptual result is progressive detail loss.

The detail anchor closes exactly this channel. Each frame is split into a spatial low band (a 3×3 average pool over the latent grid) and its high-band residual, and the mean squared energy of each band is equalised toward a *fixed* scene-first reference (captured at chunk 1, re-baselined on scene change) with symmetric per-chunk gains:

$$g_{\text{lo}} = \left(\frac{E_{\text{lo}}^{\text{ref}}}{E_{\text{lo}}} \right)^{1/2} \text{ clamped to } [0.90, 1.10], \quad g_{\text{hi}} = \left(\frac{E_{\text{hi}}^{\text{ref}}}{E_{\text{hi}}} \right)^{1/2} \text{ clamped to } [0.90, 1.12] \quad (7)$$

Unlike the shrinkage EMA, the reference here does not track the output — drift cannot legitimise itself — and unlike the shrinkage gains, these are symmetric: lost high-band energy is restored, not merely excess removed. The per-chunk clamp bounds the correction so a genuine scene change cannot be fought harder than $\pm 10\text{--}12\%$ per chunk, and a $\pm 0.5\%$ dead-band makes the module a strict no-op when energies already match. The same mechanism runs per-window in Stage 2, referenced there to the upscaled Stage-1 slice of the same window (Section 5.2). In the reported runs it holds the Stage-1 high-frequency share flat end-to-end ($0.510 \rightarrow 0.510$ I2V, $0.512 \rightarrow 0.520$ T2V; Section 6).

3.6 Stability Analysis

The AdaIN blend (Section 3.2) and context re-noising (Section 3.1) together reduce the magnitude of any per-chunk latent perturbation by a factor $(1 - \beta)(1 - \tau_c)$ before it reaches the next chunk’s denoiser. This gives the *loop gain*:

$$\rho_{\text{loop}} = (1 - \beta) \cdot (1 - \tau_c) \quad (8)$$

The combined closed-loop gain, accounting for the model’s open-loop amplification g , is:

$$\rho_{\text{closed}} = g \cdot \rho_{\text{loop}} \quad (9)$$

Bounded drift requires $\rho_{\text{closed}} < 1$. With $\beta = 0.40$ and $\tau_c = 0.05$, $\rho_{\text{loop}} = 0.57$, providing $\approx 43\%$ gain reduction per chunk before accounting for g .

Why the two factors multiply. $(1-\tau_c)$ and $(1-\beta)$ act on disjoint tensor regions at different points in the cycle — $(1-\tau_c)$ on the *overlap* region when it is *read* as context (Section 3.1), $(1-\beta)$ on the *new-frame* region when it is *written* as output (Section 3.2, and see the corrected notation there: AdaIN never re-touches the overlap). This is not two attenuators applied to the same signal at the same place; it is two attenuators the same error signal passes through exactly once per loop iteration: a perturbation in chunk N ’s output is written with gain $(1-\beta)$ (since AdaIN is applied uniformly across L_N^{new} , including the tail frames that become the next SLB), then read back with gain $(1-\tau_c)$ once it becomes chunk $N+1$ ’s overlap context. For a linear, time-invariant loop this is exactly the standard construction of a loop gain from attenuators in series — the product is valid *because* each attenuator sits on the one path the error must traverse per cycle, not because they act on the same coefficients. What Eq. (8) does not capture is any cross-term where the correction strength itself depends on chunk content (e.g. the AdaIN amplification cap, Section 3.2, engages non-uniformly); that content-dependence is exactly why g cannot be folded into a single constant either (below).

Note that Eq. (8) is a first-order approximation that treats the two corrections as independent multiplicative attenuators. In practice g is not constant: it depends on the step count, guidance strength, noise schedule, and the spectral properties of the current latent. The analysis therefore provides a stability condition rather than a precise quantitative bound. The frequency-band shrinkage (Section 3.3) provides additional spectral attenuation beyond ρ_{loop} , especially in high-frequency bands, but is not included in the linear gain model because its contribution is content- and chunk-dependent.

Hyperparameter Choices. $\beta = 0.40$ is the largest value that preserves scene-change responsiveness in practice: at $\beta \geq 0.5$ slow colour shifts accumulate due to excessive EMA anchoring. $\tau_c = 0.05$ is a conservative re-noising level that maintains high boundary similarity (> 0.97) while partially correcting SLB distribution shift; larger values ($\tau_c \geq 0.20$) cause visible motion discontinuities on 22B streaming. Together they yield $\rho_{\text{loop}} = 0.57$, which empirically provides adequate margin given the band-energy-growth proxy for g discussed in Section 7 (note: not the g_{SLB} perturbation measurement below, which was not exercised in the reported runs). $\lambda = 0.10$ (EMA rate) was chosen to respond to gradual scene evolution without over-weighting individual chunks; $\gamma = (0.0, 0.05, 0.2, 0.3)$ are conservative exponents that attenuate only bands with clear excess energy.

Per-Chunk Gain Measurement. g is measured empirically by running a second denoising pass with a slightly perturbed overlap latent $\tilde{L}_{\text{overlap}} + \delta$ ($\|\delta\|_F = \epsilon \|\tilde{L}_{\text{overlap}}\|_F$, $\epsilon = 0.01$) and comparing the output perturbation to the input:

$$g_{\text{SLB}} = \frac{\|\Delta_{\text{output}}^{\text{SLB}}\|_F}{\|\delta\|_F} \quad (10)$$

where $\Delta_{\text{output}}^{\text{SLB}}$ is restricted to the last overlap/new frames—the portion that becomes the next chunk’s SLB. This measures sensitivity of the boundary output to boundary

input perturbations, which is the relevant quantity for SLB-mediated drift propagation. It does not capture drift arising from other channels (e.g., prompt conditioning or attention to non-overlap frames), so g_{SLB} is a lower bound on the true open-loop gain.

The `measure_g` config field existed from the start, but — caught only when this was checked directly rather than assumed — no code actually implemented Eq. (10) behind it; the flag was a documented no-op, hardcoded off with no node-level toggle. This has since been implemented and exposed on the **CLSSConfig** node (Section 5.4): a second denoising pass with a perturbed overlap latent, holding the noise sample and every other conditioning input fixed, comparing the raw (pre-AdaIN/shrink) output tail against the unperturbed pass. It is verified at the tensor level — the perturbation is confined exactly to the overlap region, the norm-ratio formula recovers a known injected gain in a synthetic test — and was enabled for the I2V experiment reported in Section 6.3; the measured values, and their consequences for the stability model, are reported in Section 7. The $g \approx 1.03$ – 1.05 figures quoted there from earlier runs are a *different*, informal proxy — the ratio of measured band-3 latent energy growth to its EMA reference — not an instance of Eq. (10). The two should not be conflated: the band-energy proxy reflects unshrunk spectral drift under the full correction stack as actually run, while g_{SLB} isolates the transformer’s own sensitivity to a boundary perturbation, holding every correction fixed.

4 Algorithm

Algorithm 1 summarises the per-chunk CLSS procedure.

Algorithm 1 CLSS per-chunk generation

Require: Config: $\tau_c, \beta, \lambda, \gamma$; state: SLB, EMA, anchor bank

- 1: $L_{\text{ov}} \leftarrow \text{SLB.read}()$ ▷ may be empty for chunk 0
 - 2: $\tilde{L}_{\text{ov}} \leftarrow (1 - \tau_c) L_{\text{ov}} + \tau_c \varepsilon$ ▷ Eq. (1); Sect. 3.1
 - 3: $A \leftarrow \text{anchor_bank.retrieve}(\text{top-}m \text{ by cosine sim})$ ▷ Sect. 3.4
 - 4: $L_N \leftarrow \text{Denoise}(\tilde{L}_{\text{ov}}, A, \text{prompt}_N)$
 - 5: $L_N^{\text{new}} \leftarrow L_N[\text{overlap_lf} :]$ ▷ drop overlap region
 - 6: $L_N^{\text{new}} \leftarrow \text{AdaIN_blend}(L_N^{\text{new}}, \text{EMA}, \beta)$ ▷ Sect. 3.2
 - 7: $L_N^{\text{new}} \leftarrow \text{band_soft_shrink}(L_N^{\text{new}}, \text{band_EMA}, \gamma)$ ▷ Sect. 3.3
 - 8: $L_N^{\text{new}} \leftarrow \text{detail_anchor}(L_N^{\text{new}}, E_{\text{lo/hi}}^{\text{ref}})$ ▷ Eq. (7); Sect. 3.5
 - 9: $\text{EMA.update}(L_N^{\text{new}}, \lambda)$; $\text{band_EMA.update}(L_N^{\text{new}}, \lambda)$
 - 10: $\text{anchor_bank.update}(\text{last frame of } L_N^{\text{new}})$ ▷ Sect. 3.4
 - 11: $\text{SLB.push}(L_N^{\text{new}}[\text{—overlap_lf} :])$
 - 12: **return** L_N^{new}
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5 Implementation

5.1 Memory Budget for 16 GB VRAM

LTX-2.3 22B in BF16 occupies ≈ 44 GB. Three complementary strategies fit it within 16 GB VRAM:

1. **GGUF 4-bit quantisation** (Q4_K_S): reduces the on-disk size to ≈ 11 GB. De-quantization to BF16 at load time requires ≈ 44 GB of pinned CPU RAM.
2. **CPU block-streaming**: all transformer blocks reside in pinned CPU RAM; only 2 blocks are resident on GPU at a time (≈ 5 GB model weights on GPU), with the next block pre-fetched asynchronously via a dedicated CUDA stream.
3. **Temporal chunking (CLSS)**: latent memory is $O(\text{overlap_lf})$ instead of $O(\text{total_frames})$. Processed chunk latents are immediately offloaded to CPU after each chunk.

Combined peak VRAM budget at 512×768 resolution:

Component	Peak VRAM
Model blocks ($2 \times$ BF16, streaming)	≈ 5 GB
Activations + latents (per chunk)	≈ 4 GB
VAE / text encoder (loaded/freed)	≈ 2 GB
Total peak	≈ 11 GB

5.2 Two-Stage Pipeline

Stage 1 (Open-resolution streaming). CLSS streaming generates the full video at a coarse latent resolution ($H_L \times W_L$ latent grid, typically 11×20 for 704×512 output). Each chunk runs the full 20–30-step LTX scheduler. CLSS corrections are applied between chunks as per Algorithm 1.

Stage 2 (Distilled-LoRA refinement). Stage 1 operates at 704×512 output resolution and 20 denoising steps, which is sufficient for temporal structure and scene content but produces visible quantisation artefacts and reduced texture detail, particularly after GGUF 4-bit dequantisation at the low-frequency latent scale. Stage 2 addresses this by upsampling and refining the output.

The Stage 1 output is spatially upsampled $2\times$ by `LTXVLatentUpsampler`, yielding a 22×40 latent grid corresponding to 1408×1024 pixel resolution. Stage 2 then refines the full video in temporal chunks using a 3-step distilled-LoRA schedule with initial noise level $\sigma_0 \approx 0.909$. Each chunk’s new frames are seeded from the upscaled Stage 1 latent with σ_0 -scaled noise:

$$x_{\text{start}} = \sigma_0 \varepsilon + (1 - \sigma_0) L_N^{\text{S1}} \quad (11)$$

This formulation anchors Stage 2 to the coarse temporal structure produced by Stage 1—preserving scene layout, motion trajectories, and chunk boundaries established by

CLSS—while allowing the 3-step refinement to add local high-frequency detail. Stage 1 alone could in principle be generated at $2\times$ resolution, but this would quadruple token count and exceed VRAM budget under CPU block-streaming.

Key differences from Stage 1:

- AdaIN and spectral-shrinkage corrections are *disabled*: Stage 2 refinement is structurally anchored to the S1 latent, so distributional drift correction is unnecessary.
- The detail anchor (Section 3.5) *does* run, per window, with the reference re-computed from that window’s upscaled Stage-1 slice rather than carried from a first-window EMA. The 3-step distilled refinement systematically adds spatial high-frequency energy relative to its Stage-1 seed, and without the anchor this excess compounds across the window chain as end-of-run softening of the refined output relative to Stage-1 structure. Per-window re-referencing means the correction cannot accumulate: each window is equalised to its own source.
- A global pre-generated noise field is sliced per chunk (`_SlicedNoise`) to prevent grain seams at chunk boundaries—a single coherent noise tensor ensures that the per-chunk noise samples tile seamlessly when concatenated.
- Audio is *frozen*: the Stage-1 audio latent is passed through unchanged (denoise mask 0) and only video tokens are refined, with the frozen audio serving as cross-modal context. Re-rendering audio in Stage-2 windows was measured to re-seam it at every window boundary; freezing keeps the audio timeline bit-identical to Stage 1 (`frozen_verify_sim=1.0` per window).

5.3 Audio Streaming

Audio and video share the same temporal chunk boundaries. Audio length accounting is anchored to *pixel time*: the causal video VAE maps a chunk of `new_lf` latent frames to $(\text{new_lf} - 1) \cdot 8 + 1$ pixel frames for the first chunk of a sequence and $\text{new_lf} \cdot 8$ pixel frames for every continuation chunk. The audio rate is taken from the chunk template, $r = \text{new_af} / ((\text{new_lf} - 1) \cdot 8 + 1)$ audio frames per pixel frame, and the SLB length follows the same rate:

$$\text{overlap_af} = \text{round}(\text{overlap_lf} \cdot 8 \cdot r) \quad (12)$$

With `overlap_lf = 8` and the 13-lf/102-af template this gives `overlap_af = 67`. (An earlier per-chunk accounting ignored the causal first-frame discount, undercounting every continuation chunk by $\approx 0.28\text{s}$ and desynchronising audio from video by $\approx 2\text{s}$ over seven chunks; the pixel-time ledger above eliminates the drift exactly.) The effective video overlap is additionally clamped to `new_lf`: the SLB can never carry more frames than a chunk produces, and unclamped values were measured to silently discard motion at every seam.

Continuity via reference audio, not a buffer. Following the reference pipeline, audio continuity is carried by *conditioning*, not by a seeded buffer: the overlap-time audio of each continuation chunk carries an audio analogue of the video SLB: the

overlap audio frames are seeded from the previous chunk’s tail and re-noised at τ_c , so each chunk opens with near-verbatim continuity. A controlled same-seed A/B on the current pipeline shows this carrier is load-bearing: with the audio SLB enabled, audio boundary similarity holds at 0.76–0.94; with it disabled (continuity via the reference window alone) seams collapse to 0.28–0.58 and high-frequency energy runs away ($0.54 \rightarrow 1.25$ over ten chunks), audible as sudden jumps and degraded speech. The SLB is therefore standard (always on, not a user option). Its near-verbatim tail also carries the per-chunk loudness gesture forward — the repetition loop measured in Section 7.6 — so the final pipeline re-noises the audio SLB on a hotter schedule than the video SLB (τ_c rising $\approx 0.15 \rightarrow 0.28$ over the run, versus $0.05 \rightarrow 0.10$ for video) and envelope-flattens the audio context before conditioning: each context frame is rescaled toward the window-mean RMS (gain capped at $[0.6, 1.25]$, asymmetric so quiet frames are never over-amplified), removing the loudness arc while preserving content and timbre. Alongside the SLB, a reference-audio window (`ref_audio` tokens at negative temporal RoPE) taken from the previous chunk’s pre-overlap tail provides content conditioning; its length is derived from the audio overlap (67 af ≈ 2 s at the standard configuration). Other perturbations of this reference — blending in noise, shrinking its window over chunks — were tested live and reduced nothing, adding audible hiss; the clean full-length reference is the measured optimum (Section 7.6).

Reference-exact RMS anchoring. Un-anchored streaming audio is a divergent loop: runs were measured with RMS growing +67% and low-mid spectral bands tripling over ten chunks. A per-chunk anchor corrects the new audio’s global RMS toward an EMA target whose drift from the chunk-1 reference (computed onset-excluded) is hard-capped at $\pm 5\%$ (the same Δ_σ used for the video EMA), before the chunk’s tail is fed forward as the next reference window. Anchoring *before* feed-forward is what keeps the recursion safe at any sequence length. In the reported T2V run this anchor absorbs a raw loudness dip to 0.814 at chunk 2 (−18%, corrected with gain 1.20) and holds output RMS at 0.962–0.978 against a chunk-1 reference of 0.996 (Section 6.2.3). On multi-scene runs the anchor is re-baselined at scene changes; this is measurably a ratchet (each re-baseline legitimises accumulated loudness creep: $0.599 \rightarrow 0.699 \rightarrow 0.823$ over three scenes in one run) and remains a known limitation.

Boundary smoothing. At each audio chunk boundary, a linear blend over ± 2 audio latent frames is applied in latent space before VAE decoding. This eliminates click artifacts that arise from the abrupt splice between consecutive chunks, which would otherwise be visible as discontinuities in the spectrogram.

5.4 ComfyUI Node Integration

Five custom nodes implement the full pipeline inside ComfyUI:

1. **CLSSConfig**: exposes the CLSS hyperparameters worth exposing (τ_c, β , overlap); validated internals are fixed. Two diagnostic levers are also exposed: `shrinkage=off`

zeroes γ (the spectral-shrinkage ablation arm), and `measure_g` (with `measure_g_epsilon`) enables the g_{SLB} measurement of Eq. (10) at $\approx 2\times$ Stage-1 compute.

2. **CLSSScenePrompts**: multi-scene text input via `---` delimiters; Gemma-enhanced per scene. Output is a flat CONDITIONING list with one entry per scene, proportionally assigned to chunks. With one prompt per chunk this doubles as a per-chunk *object-identity* anchor (see Section 7.2).
3. **CLSSStreamingSampler**: Stage 1 streaming sampler. Accepts any ComfyUI GUIDER/SAMPLER/SIGMAS/NOISE/LATENT combination. Emits an unconditional settings dump, per-chunk telemetry, end-of-run trend summaries with per-chunk localisation, per-frame origin/layout drift event tables, and a phase-lock repetition detector. Two experiment levers widen this: `open_loop=on` disables every closed-loop correction (re-noising, AdaIN, shrinkage, detail and std anchors, audio RMS anchor) to give the naive sliding-window baseline arm, and `object_identity=on` adds a spatially-resolved identity metric (`vid_obj`) that the mean-pooled `identity_sim` cannot provide (Section 7.2).
4. **CLSSStage2**: Stage 2 refinement sampler with consistent per-chunk noise slicing, per-window detail anchoring, and frozen-audio passthrough (Section 5.2).
5. **CLSSAVGuiderV2**: reference-parity multi-modal guider (Section 5.5). Its predecessor (**CLSSAVGuider**, a `sampler_cfg_function` patch) is retained but inert on current ComfyUI and warns accordingly.

All stochastic inputs are seeded: Stage-1 video *and* audio noise are pre-generated as full-sequence fields sliced per chunk (distinct yet coherent noise per chunk, bit-reproducible runs), and every fallback noise path uses a generator derived deterministically from the seed and chunk position.

5.5 Multi-Modal Guidance and the Packed-Latent Pitfall

The reference pipeline guides each modality independently (**MultiModalGuider**):

$$\text{pred} = c + (s_{\text{cfg}} - 1)(c - u) + s_{\text{stg}}(c - p) + (s_{\text{mod}} - 1)(c - m) \quad (13)$$

per modality, followed by a rescale $\text{pred} *= r \sigma_c / \sigma_{\text{pred}} + (1 - r)$, where u is the text-unconditional prediction, m the modality-isolated prediction (both audio $\leftrightarrow$ video cross-attentions skipped), and p the STG-perturbed prediction. The STG perturbation skips *self*-attention in designated blocks (block 28 for LTX-2.3) by replacing the attention output with the raw value projection ($\text{out} = W_V x$, gating and output projection unchanged) — a value-passthrough, not a zero-out. The reference defaults are video cfg 4–4.5, audio cfg 7, modality 3, STG 1 for both modalities, rescale 0.7 (four transformer passes per step). The working configuration runs $s_{\text{mod}} = 1$: the modality term of Eq. (13) vanishes and its pass is skipped (three passes per step). This is a measured choice, not an economy: raising modality guidance to the reference value 3 was tested live and made the audio audibly *worse* (hiss/drone; within-chunk audio coherence collapsed $0.68 \rightarrow 0.61$) while leaving the audio repetition it might have addressed unchanged — with Stage-2 audio frozen, the raw guided Stage-1 audio is the final audio, and additional guidance energy surfaces directly as audible high-frequency content.

The packed-latent representation. ComfyUI’s `CFGGuider.sample()` packs nested audio-video latents into a *single flat* $[B, 1, N_v + N_a]$ tensor before sampling and unpacks only after. A guider that tries to detect the audio-video case with `isinstance(x, NestedTensor)` inside `predict_noise` therefore never sees a nested tensor and falls back to plain shared CFG — and it does so silently, producing no error and no log line. The guider must instead capture the per-modality shapes in `sample()` *before* packing, detect the packed $[B, 1, N]$ representation in `predict_noise`, unpack each pass output, apply Eq. (13) per modality, and repack. Two diagnostics guard against a silent regression: the guider announces the active branch and its pass count once per chunk (`passes/step=3` for the working configuration, 4 with modality guidance enabled), and it logs the STG difference norms so a zero-valued (inert) perturbation is visible immediately. A useful invariant follows from the pass count: because each guidance term costs a transformer pass, a guidance configuration that does not change per-step wall-clock time is not actually running — in an earlier 13-lf configuration the four-pass stack cost $\approx 17.3\text{s/it}$ against $\approx 8.0\text{s/it}$ for plain CFG. Two engineering rules generalise: guidance paths must announce which branch is active, loudly and unconditionally; and a guidance change that does not change step time did not happen.

6 Experiments

Two generation experiments share the same CLSS configuration, seed, and active guidance stack but differ in the first-frame conditioning: a text-to-video (T2V) run with no image prior (Section 6.2), and an image-to-video (I2V) run conditioned on a single guide frame ($1 \times 11 \times 20$ latent, Section 6.3). Both stream ten Stage-1 chunks (`new_lf = 16`, `overlap = 8`) to 160 latent frames ($\approx 53\text{s}$ at 24 fps) and refine through the same four-window Stage 2 pass.

6.1 Shared Configuration

Table 1: Shared configuration. Guidance values are those actually active during sampling (verified by the guider’s per-chunk `passes/step` and difference-norm telemetry, Section 5.5). The Stage-2 geometry is selected automatically from the VRAM token budget.

Parameter	Value
Model	LTX-2.3 22B (Q4_K_S GGUF)
Resolution	704×512 (Stage 1), 1408×1024 (Stage 2)
τ_c	0.05
β	0.40
λ	0.10
Δ_σ	0.05
F_a	3 (auto, for 10 chunks)
γ	(0.0, 0.05, 0.2, 0.3)
Detail anchor	on (Stage 1 scene-first ref; Stage 2 per-window ref)
Video guidance	CFG 4.0, STG 1.0
Audio guidance	CFG 7.0, STG 1.0
Modality guidance	1.0 (disabled; Section 5.5)
STG block / rescale	28 / 0.70 (three passes/step)
S1 noise temporal-corr mix	0.30 (video only)
Audio continuity	<code>ref_audio</code> 67 af; audio SLB 67 af, τ_c 0.15 $\rightarrow$ 0.28 sched., env-flattened c
Audio anchor	RMS-only, reference-exact $\pm 5\%$
Stage 1 steps	20
Stage 2 steps	3 (distilled), audio frozen (mask 0)
Diagnostics	<code>object_identity=on</code> (<code>vid_obj</code>); g_{SLB} measured on a same-seed compar
VRAM (GPU)	15.61 GB (RTX-class)
T2V first-frame	Text only
I2V first-frame	Guide image, latent [1, 128, 1, 11, 20]
Stage 1 (both)	10 chunks, <code>new_lf</code> = 16, <code>overlap</code> = 8, 1 scene
Stage 2 (both)	4 windows, <code>fpc</code> = 40, <code>overlap</code> = 13 (auto)

Ablation and measurement levers. Each correction, and the open-loop baseline itself, maps to a single node toggle, so every arm of an ablation matrix is a one-click variant of the reported configuration: the naive open-loop baseline (`open_loop=on`); re-noising off ($\tau_c = 0$); AdaIN off ($\beta = 0$); spectral shrinkage off (`shrinkage=off`); detail anchor off; audio RMS anchor off; and the guidance terms individually (modality scale 1.0, STG 0, rescale 0 on **CLSSAVGuiderV2**). The g_{SLB} measurement (Eq. (10)) is enabled by `measure_g=on`. These ablation and measurement runs are planned follow-up work; no results for these arms are reported here.

6.2 Experiment 1: Text-to-Video (T2V)

No image is provided; Stage 1 chunk 1 begins from pure noise conditioned on the text prompt alone. The clip is a single prompted scene streamed as ten chunks ($\text{new_lf} = 16$, $\text{overlap} = 8$), assembling to 160 latent frames ($\approx 53\text{s}$), under the identical seed and guidance stack as the I2V run (Table 1). The prompted content is the more dynamic of the two arms: intra-chunk similarity (a chunk’s first vs. last new frame) fluctuates between 0.13 and 0.88 with no settling plateau (Table 3), where I2V’s rises to a 0.86–0.94 plateau, so this run exercises the correction stack under sustained motion rather than a stable anchored scene.

6.2.1 Guidance Activity

Every chunk shows `passes/step=3` with non-zero STG difference norms on both modalities (Table 2); no step fell back to plain shared CFG. With no guide image, chunk 1 is unremarkable on the video side (83.6, the smallest of the run); the largest video response is chunk 6’s 216.7, against 94.9–172.0 otherwise. The audio STG norm climbs over the run ($15.3 \rightarrow 32.3$) as the raised audio re-noise schedule leaves more of the overlap to be reconstructed — the same pattern as I2V (Table 7).

Table 2: T2V Stage 1 guidance difference norms (step 1 of each chunk). Modality guidance is disabled ($s_{\text{mod}} = 1$), so only the STG terms are reported.

Chunk	STG (vid)	STG (aud)
1	83.6	15.3
2	94.9	28.3
3	118.7	26.9
4	140.5	28.2
5	139.0	31.6
6	216.7	38.2
7	132.1	30.4
8	169.8	31.4
9	147.7	28.1
10	172.0	32.3

6.2.2 Stage 1 Results

Table 3: T2V Stage 1 per-chunk video metrics (10 chunks). Boundary_sim compares the last overlap frame of the previous chunk against the first new frame; identity_sim compares the chunk against its nearest anchor-bank entry; vid_obj is the grid-cell cosine (8×8 , decoded seam frames) mean with the per-cell minimum in parentheses; g_{SLB} is the open-loop gain per Eq. (10), measured in this run itself (measure_g=on, $\epsilon = 0.01$).

Chunk	Boundary sim $\uparrow$	Intra sim $\uparrow$	Identity sim	vid_obj	g_{SLB}	ρ_{loop}
1/10	N/A	0.4802	1.0000 (ref)	—	—	0.570
2/10	0.8553	0.4019	0.8553	0.955 (0.762)	87.0	0.570
3/10	0.7915	0.6426	0.6547	0.986 (0.905)	28.1	0.570
4/10	0.8576	0.1286	0.8576	0.968 (0.840)	48.6	0.570
5/10	0.8513	0.8819	0.8513	0.996 (0.979)	29.0	0.570
6/10	0.8174	0.6224	0.7492	0.994 (0.954)	30.6	0.570
7/10	0.8456	0.8219	0.8219	0.994 (0.963)	15.8	0.570
8/10	0.7479	0.4951	0.7671	0.991 (0.952)	18.8	0.570
9/10	0.8443	0.7970	0.8829	0.995 (0.968)	29.7	0.570
10/10	0.9831	0.6548	0.9230	0.990 (0.920)	23.6	0.570

Video boundary continuity holds between 0.748 and 0.983 across the nine chunk transitions, with no early-chunk deficit: the floor sits mid-run at chunk 8 (0.7479, the trend summary’s boundary WARN) and the final seam is the best of the run (0.9831) — the corrected loop does not compound error over this horizon. The largest structural events still cluster at the chunk 1→2 hand-off: four of the six largest per-frame origin-similarity drops land in chunk 2 ($t = 5.3$ – 8.7 s, deepest $\Delta = -0.2385$ at $t = 5.7$ s), and the largest coarse-layout drops split between chunk 2 (four events, deepest $\Delta = -0.3452$) and chunk 4 (two). Origin similarity to the scene reference trends downward over the run ($0.261 \rightarrow 0.109$, flagged as drift) as the moving content walks away from its opening, while identity similarity to the anchor bank holds 0.655–0.923 (floor at chunk 3) with no monotonic decay. Intra-chunk similarity fluctuates (0.129 at chunk 4 to 0.882 at chunk 5) — sustained motion within every chunk, the complementary regime to I2V’s anchored plateau. Chunk 2’s identity similarity equals its boundary similarity exactly (0.8553): with a single-entry bank the nearest anchor *is* the previous chunk’s tail, so the two measurements coincide — the anchor/SLB self-consistency property. The decoded-frame object-identity grid confirms the seam frames carry the scene: cell-cosine means hold 0.955–0.996 with worst cell 0.762 (chunk 2). The measured open-loop gain g_{SLB} reads 15.8–87.0 per chunk (max at chunk 2, decaying to 23.6 by chunk 10) — an order of magnitude above the run’s own stability-ceiling line, which requires $g < 1.754$ for $\rho_{\text{closed}} < 1$ at $\rho_{\text{loop}} = 0.570$, at every measured seam.

Table 4: T2V Stage 1 AdaIN corrections, band gains, and detail-anchor gains (g_{lo}/g_{hi} , Eq. (7); Band 0 is the DC band, $\gamma_0 = 0$, $gain \equiv 1$, omitted).

Chunk	$\Delta\mu$	$\Delta\sigma$	g_1	g_2/g_3	g_{lo}	g_{hi}
1	+0.0000	+0.0000	1.0000	1.0000/1.0000	ref	ref
2	−0.0006	−0.0077	0.9952	0.9955/0.9959	0.9978	1.0077
3	−0.0058	−0.0339	0.9936	0.9776/0.9812	1.0489	0.9384
4	−0.0063	−0.0299	0.9932	0.9835/0.9937	1.0157	0.9660
5	−0.0042	−0.0557	0.9932	0.9716/0.9846	1.0059	0.9189
6	+0.0016	−0.0238	0.9950	0.9890/0.9987	1.0138	0.9630
7	−0.0014	−0.0283	0.9955	0.9902/0.9975	1.0180	0.9556
8	+0.0010	−0.0279	0.9916	0.9922/0.9978	0.9938	0.9596
9	−0.0069	−0.0302	0.9963	0.9921/0.9962	1.0016	0.9702
10	−0.0046	−0.0415	0.9949	0.9786/0.9812	0.9927	0.9295

Every T2V continuation chunk receives a *downward* std correction ($\Delta\sigma < 0$ throughout, to -0.056 at chunk 5): each text-only chunk’s raw std runs hot relative to the EMA reference (pre-AdaIN std up to 1.057) and AdaIN pulls it back to ≈ 0.95 . The corrections stay moderate — much closer to I2V’s every-chunk trim (Table 9) than to a fight at the caps: band-2 gains bottom out at 0.9716, and the detail anchor never reaches its 0.90 floor cap. Raw high-band energy still overshoots the scene-first reference on every chunk from chunk 3 (the model keeps adding spatial detail energy under sustained motion), and g_{hi} trims it back to 0.919–0.970 while g_{lo} stays within 5% of unity. The net effect is a high-frequency *share* held exactly flat at 0.506 over the full run (trend range 0.000), matching I2V’s flatline.

The anchor bank grows identically to I2V, from 1 to 5 entries at the same frame positions (`bank_frame_ids` = [15, 47, 63, 111, 159]). With `new_lf` = 16, chunk k ’s last frame lands at absolute index $16k - 1$: 15, 31, 47, 63, ... for $k = 1, 2, 3, \dots$ — so these five entries are chunks {1, 3, 4, 7, 10}, not chunks 2, 5, 6, 8, or 9. Four of the five are the forced-insertion schedule exactly (`chunk_index mod  $F_a$` = 0 with $F_a = 3$: chunk 1 always seeds, then chunks 4, 7, 10). Chunk 3 is the one *organic*, threshold-triggered commit of the run: chunk 2’s similarity to the seed anchor was already below threshold (`last_frame_max_sim` = 0.5045, `streak` = 1), and chunk 3 pushed the streak to $P = 2$, committing independently of the forcing schedule. Because the forced positions are fixed by chunk index alone, and T2V and I2V share chunk boundaries and seed, the two runs’ forced entries coincide exactly despite entirely different video content; only an organic (content-triggered) entry could differ between them, and none did here (I2V’s chunk 2 sits at 0.0118 against the same threshold — the organic commit lands at the same chunk in both runs).

6.2.3 Stage 1 Audio

Table 5: T2V Stage 1 per-chunk audio metrics. Within-chunk similarity is reported at the start and end of each chunk; env_corr is the loudness-envelope correlation with the previous chunk (> 0.7 flags a repeating gesture); RMS raw/out are pre-/post-anchor (chunk-1 onset-excluded reference 0.996).

Chunk	WC start	WC end	env_corr	RMS raw	RMS out	Bnd sim
1	0.800	0.669	—	0.993	(ref)	N/A
2	0.501	0.397	0.174	0.814	0.978	0.9106
3	0.683	0.471	0.223	0.901	0.971	0.8938
4	0.819	0.508	0.347	0.910	0.965	0.8542
5	0.869	0.650	0.239	0.940	0.962	0.4340
6	0.853	0.719	0.176	0.990	0.965	0.9612
7	0.886	0.725	0.198	0.999	0.968	0.9641
8	0.720	0.846	0.267	0.968	0.968	0.8313
9	0.772	0.814	0.316	0.994	0.971	0.9203
10	0.946	0.799	0.212	0.970	0.971	0.9502

As in I2V, the envelope repetition is gone: with the audio context envelope-flattened (per-chunk flatten gains spanning $[0.693, 1.250]$), the loudness-envelope correlation with the previous chunk never crosses the 0.7 repetition flag — it reads 0.174, 0.223, 0.347, 0.239, 0.176, 0.198, 0.267, 0.316, 0.212 over chunks 2–10. The energy-peak positions do not phase-lock either (frames 98, 68, 125, 39, 97, 99, 22, 31, 88, 2; the mode covers only 2/10 chunks), and the coarse-layout argmin is likewise unclustered (mode covers 4/10). Loudness stays near reference and the RMS-only anchor trims it gently: raw per-chunk RMS dips to 0.814 at chunk 2 (corrected with gain 1.20) and otherwise wanders between 0.90 and 1.00, while the anchored output holds 0.962–0.978 against the 0.996 reference. The high-frequency share *rises* over the run ($0.607 \rightarrow 0.835$, trend range 0.227) and the top frequency bin reaches $1.34\times$ the chunk-1 reference by chunk 10. Audio boundary similarity is high throughout (0.83–0.96) with one exception: chunk 5’s seam drops to 0.4340 and recovers immediately; within-chunk similarity floors at 0.397 at chunk 2 (both flagged WARN by the trend summary) and ends at 0.799. With the audio SLB on (auto) under the raised re-noise schedule ($\tau_{c,\text{eff}} = 0.150 \rightarrow 0.284$, base $\times 3$, ceiling 0.35), the 67-frame overlap is re-noised ever more aggressively yet reconstructed essentially exactly — SLB honoured ≥ 0.9999 on every chunk — so continuity across the cut is enforced directly, not left to the reference-audio conditioning alone.

6.2.4 Stage 2 Results

Stage 2 refines the $2\times$ -upsampled latent in four overlapping windows ($\text{fpc} = 40$, overlap = 13, auto-selected from the 42k-token VRAM budget) against frozen Stage-1 audio (Section 5.2).

Table 6: T2V Stage 2 per-window metrics (4 windows). g_{lo}/g_{hi} are the per-window detail-anchor gains (reference: the window’s own upscaled Stage-1 slice); Aud_bnd is the frozen Stage-1 audio’s own continuity at the window seam (Stage 2 never re-rolls audio).

Window	Bnd sim $\uparrow$	Intra sim	Fid first	Fid last	Aud_bnd	g_{lo}/g_{hi}
1/4	N/A	0.2145	0.9220	0.9094	N/A	1.074/0.900
2/4	0.8482	0.8566	0.9657	0.9302	0.8285	1.050/0.900
3/4	0.6771	0.6212	0.9364	0.9223	0.7784	1.034/0.900
4/4	0.8293	0.6799	0.9311	0.8046	0.9562	1.071/0.900

Window 1’s intra similarity (0.215) is the lowest of the four windows: the dynamic T2V opening (Table 3) carries into Stage 2 as a first window whose start and end frames correlate only loosely, whereas I2V’s guide image keeps its first window internally coherent (0.767). First-frame fidelity to the Stage-1 latent holds 0.92–0.97 across all windows (0.922 $\rightarrow$ 0.931; the trend WARN fires on the per-window minimum, not a decline), but end-of-window fidelity erodes over the run: 0.909 at window 1 falling to 0.805 at window 4 (trend-summary WARN) — as in I2V, refinement drifts furthest from the Stage-1 input late in the clip. Video seam continuity ranges 0.68–0.85, weakest at the window-2 $\rightarrow$ 3 seam (0.6771, $t = 26.7$ s). The per-window detail anchor engages identically at every window: the 3-step distilled refinement adds spatial high-frequency energy over its Stage-1 seed (pre-anchor hf share 0.50–0.51 against Stage-1 references of 0.37–0.39), and g_{hi} rides its 0.90 floor cap in all four windows, pulling the excess back toward the Stage-1 reference; because the reference is recomputed per window, the correction cannot compound. The audio timeline is bit-identical to Stage 1 (`frozen_verify_sim`= 1.0 per window); the Aud_bnd column re-measures the Stage-1 audio’s own continuity where the video windows cut (0.828/0.778/0.956 at $t = 13.33/26.67/40.00$ s, flagged WARN on the 0.778 minimum). The assembled refined latents are [1, 128, 160, 22, 40] (video, std 0.887) and [1, 8, 1333, 16] (audio, frozen), with no NaN/Inf.

6.3 Experiment 2: Image-to-Video (I2V)

A single guide image is encoded to a latent of shape [1, 128, 1, 11, 20] and provided as first-frame conditioning to Stage 1 chunk 1 (guide adherence 1.0000 measured at the placed frame). The clip is a single prompted scene streamed as ten chunks (`new_lf` = 16, `overlap` = 8), assembling to 160 latent frames (≈ 53 s). All CLSS hyperparameters and the guidance stack are those of Table 1.

6.3.1 Guidance Activity

The guider emits `passes/step`=3 and its STG difference norms once per chunk (Table 7); both modalities stay well away from zero on every chunk, so no step fell back to plain shared CFG. The largest *video* STG responses are chunk 1 (199.5) and chunk 9

(193.5), against 83.4–155.7 otherwise; at chunk 1 the guide image is a hard constraint the perturbed prediction cannot honour, so the perturbation response is largest exactly where the conditioning is strongest. The *audio* STG norm climbs over the run ($11.3 \rightarrow 24.3$) as the raised audio re-noise schedule leaves more of the overlap to be reconstructed.

Table 7: I2V Stage 1 guidance difference norms (step 1 of each chunk). STG measures the self-attention-skip perturbation; non-zero $\Rightarrow$ guidance active. Modality guidance is disabled ($s_{\text{mod}} = 1$).

Chunk	STG (vid)	STG (aud)
1	199.5	11.3
2	83.4	13.4
3	103.4	14.8
4	142.4	18.5
5	150.5	21.0
6	125.9	20.1
7	133.7	18.2
8	116.6	19.2
9	193.5	23.5
10	155.7	24.3

6.3.2 Stage 1 Results

Table 8: I2V Stage 1 per-chunk video metrics (10 chunks). `Boundary_sim` compares the last overlap frame of the previous chunk against the first new frame; `identity_sim` compares the chunk against its nearest anchor-bank entry; `vid_obj` is the grid-cell cosine (8×8 , decoded seam frames) mean with the per-cell minimum in parentheses; g_{SLB} is the measured open-loop gain per Eq. (10), taken from a same-seed companion run (`measure_g=on`) under the previous audio SLB schedule (τ_c ceiling 0.15, no context flattening) — re-measurement under the final schedule is pending.

Chunk	Boundary sim $\uparrow$	Intra sim $\uparrow$	Identity sim	vid_obj	g_{SLB}	ρ_{loop}
1/10	N/A	0.3357	1.0000 (ref)	—	—	0.570
2/10	0.9126	0.0424	0.9126	0.952 (0.836)	39.6	0.570
3/10	0.8350	0.8990	0.8990	0.991 (0.961)	24.6	0.570
4/10	0.9623	0.9028	0.9623	0.991 (0.923)	22.0	0.570
5/10	0.9810	0.8568	0.9810	0.986 (0.874)	34.3	0.570
6/10	0.9782	0.9292	0.9404	0.989 (0.946)	25.1	0.570
7/10	0.9805	0.6863	0.9204	0.985 (0.892)	19.5	0.570
8/10	0.9406	0.9406	0.9406	0.997 (0.978)	20.3	0.570
9/10	0.9630	0.9080	0.9317	0.998 (0.979)	36.7	0.570
10/10	0.9752	0.9120	0.9326	0.995 (0.963)	19.6	0.570

Video boundary continuity holds between 0.835 and 0.981 across all nine chunk transitions. The floor sits early at chunk 3 (0.8350), continuity recovers immediately, and the final four seams all hold 0.94–0.98 — the corrected loop does not compound error over this horizon. The largest structural events of the whole run still cluster at the chunk 1→2 seam ($t \approx 5.3$ – 7.7 s): five of the six largest per-frame origin-similarity drops land there, the transition out of the guide-image-locked opening (the deepest single drop, $\Delta = -0.188$, is at $t = 3.7$ s inside chunk 1). Per-frame origin similarity stabilises at 0.24–0.34 from chunk 2 onward, though the trend summary still flags the first-to-last rise ($0.236 \rightarrow 0.305$, range 0.106) as drift — far from the monotonic walk-away seen at wider overlaps in earlier open-loop testing. Identity similarity to the anchor bank holds 0.899–0.981 (floor at chunk 3), and intra-chunk similarity *grows* from 0.336 to a 0.86–0.94 plateau from chunk 3 onward (max 0.941 at chunk 8) as the scene settles, albeit with deep dips at chunks 2 and 7 (0.042, 0.686) — a stable anchored scene, the complementary regime to T2V’s sustained motion. Chunk 2’s identity similarity equals its boundary similarity exactly (0.9126): with a single-entry bank the nearest anchor *is* the previous chunk’s tail, so the two measurements coincide — the anchor/SLB self-consistency property. The decoded-frame object-identity grid confirms the seam frames carry the scene: cell-cosine means hold 0.952–0.998 with worst cell 0.836 (chunk 2), and the measured open-loop gain g_{SLB} reads 19.5–39.6 per chunk on the companion run (max at chunk 2, decaying to 19.6 by chunk 10).

Table 9: I2V Stage 1 AdaIN corrections, band gains, and detail-anchor gains (Band 0 is the DC band, $\gamma_0 = 0$, gain $\equiv 1$, omitted).

Chunk	$\Delta\mu$	$\Delta\sigma$	g_1	g_2/g_3	g_{lo}	g_{hi}
1	+0.0000	+0.0000	1.0000	1.0000/1.0000	ref	ref
2	−0.0063	−0.0141	0.9960	0.9828/0.9956	1.0103	0.9655
3	−0.0048	−0.0120	0.9951	0.9848/0.9926	1.0609	0.9531
4	−0.0001	−0.0098	0.9948	0.9898/0.9968	1.0506	0.9653
5	−0.0002	−0.0091	0.9949	0.9899/0.9980	1.0535	0.9696
6	−0.0088	−0.0537	0.9911	0.9690/0.9804	1.0208	0.9125
7	−0.0051	−0.0161	0.9944	0.9865/0.9922	1.0470	0.9589
8	+0.0044	−0.0311	0.9943	0.9788/0.9910	1.0394	0.9338
9	+0.0033	−0.0137	0.9954	0.9904/0.9952	1.0388	0.9570
10	+0.0003	−0.0251	0.9940	0.9802/0.9918	1.0289	0.9395

On this stable anchored scene the correction stack works gently, as designed: AdaIN deltas stay within $|\Delta\mu| \leq 0.009$, $|\Delta\sigma| \leq 0.054$, every band gain is ≥ 0.969 , and the detail-anchor gains never stray more than 6.1% (low band) / 8.8% (high band) from unity — an every-chunk trim, not a fight. The result is the flattest headline telemetry of either run: high-frequency share exactly flat ($0.498 \rightarrow 0.498$, trend range 0.000), std stable ($0.961 \rightarrow 0.970$). T2V tells the same story at a slightly hotter setting (Table 4): band gains bottom at 0.9716 and the detail anchor never reaches its 0.90 floor — in the final pipeline neither mode fights at the caps, and the

corrections scale with measured excess rather than applying a fixed schedule.

The anchor bank grows from 1 to 5 entries over the run at chunks $\{1, 3, 4, 7, 10\}$ (`bank_frame_ids = [15, 47, 63, 111, 159]`) — the same forced-insertion-plus-one-organic-commit pattern derived in detail for T2V (Section 6.2), since both runs share chunk boundaries and seed.

The ComfyUI sampler uses the bank for identity telemetry only: the retrieval-injection conditioning path of Section 3.4 is present in the reference library but not wired into this sampler, so the bank measures identity drift without correcting it. Two conditioning wirings have been tried and both failed (Section 7.6): the reference library’s own `VideoConditionByKeyframeIndex` appends a separate token block and reproduces the audio corruption documented for the i2v guide path (Section 5.5); an in-place low-strength replacement at the first new frame instead produced visible content morphing: anchor retrieval returns the most-similar *non-redundant* anchor, and the redundancy filter excludes precisely those anchors that match the current overlap, so the retrieved frame is guaranteed to differ from the content the model is continuing — injecting it drags the chunk back toward earlier scene content. No safe anchor conditioning mechanism is currently known; the bank remains diagnostic-only.

6.3.3 Stage 1 Audio

Table 10: I2V Stage 1 per-chunk audio metrics. Within-chunk similarity is reported at the start and end of each chunk; `env_corr` is the loudness-envelope correlation with the previous chunk (> 0.7 flags a repeating gesture); RMS raw/out are pre-/post-anchor (chunk-1 onset-excluded reference 0.622).

Chunk	WC start	WC end	env_corr	RMS raw	RMS out	Bnd sim
1	0.376	0.684	—	0.619	(ref)	N/A
2	0.576	0.619	0.248	0.578	0.618	0.8216
3	0.604	0.563	0.173	0.607	0.616	0.7639
4	0.583	0.610	0.328	0.636	0.618	0.9139
5	0.640	0.647	−0.141	0.655	0.622	0.9437
6	0.638	0.667	0.187	0.658	0.626	0.8063
7	0.681	0.681	0.205	0.682	0.631	0.8251
8	0.686	0.579	0.559	0.670	0.635	0.8020
9	0.657	0.557	0.319	0.667	0.638	0.9433
10	0.821	0.715	0.077	0.682	0.643	0.8559

The headline of this run is that the envelope repetition is gone. With the audio context envelope-flattened (always on, context only; per-chunk flatten gains spanning $[0.69, 1.25]$), the loudness-envelope correlation with the previous chunk never crosses the 0.7 repetition flag: it reads 0.248, 0.173, 0.328, −0.141, 0.187, 0.205, 0.559, 0.319, 0.077 over chunks 2–10, and the first-to-last trend is *falling* ($0.248 \rightarrow 0.077$). The energy-peak positions no longer phase-lock (frames 38, 27, 84, 18, 81, 78, 129, 130, 78;

the mode covers only 3/10 chunks), in contrast to the locked late-run repetition of the unflattened pipeline whose failure modes are analysed in Section 7.6. Loudness drifts only mildly and the RMS-only anchor trims it gently: raw per-chunk RMS wanders between 0.578 and 0.682 (at most +9.6% over the reference), while the anchor holds the output at 0.616–0.643 with corrections of at most 7.4% ($g = 0.926$ at chunk 7). The spectral balance does not grow: the high-frequency share *falls* over the run ($0.974 \rightarrow 0.897$, trend range 0.207), and the top-bin energy ratios reach only $\approx 1.8\times$ the reference by chunk 10. Audio boundary similarity is high throughout (0.764–0.944, floor at chunk 3): with the audio SLB on (auto) under the raised re-noise schedule ($\tau_{c,\text{eff}} = 0.150 \rightarrow 0.284$, base $\times 3$, ceiling 0.35), the 67-frame overlap is re-noised ever more aggressively yet reconstructed essentially exactly — SLB honoured ≥ 0.9998 on every chunk — so continuity across the cut is enforced directly, not left to the reference-audio conditioning alone.

6.3.4 Stage 2 Results

Stage 2 refines the $2\times$ -upsampled latent in four overlapping windows (fpc = 40, overlap = 13, auto-selected from the 42k-token VRAM budget) against frozen Stage-1 audio (Section 5.2).

Table 11: I2V Stage 2 per-window metrics (4 windows). Fid is video fidelity to the Stage-1 latent (first / last frame of the window); Aud_bnd is the frozen Stage-1 audio’s own continuity at the window seam; $g_{\text{lo}}/g_{\text{hi}}$ are the per-window detail-anchor gains.

Window	Bnd sim $\uparrow$	Intra sim	Fid first	Fid last	Aud_bnd	$g_{\text{lo}}/g_{\text{hi}}$
1/4	N/A	0.7670	0.9834	0.9401	N/A	1.050/0.900
2/4	0.8540	0.8227	0.9473	0.9485	0.5921	1.056/0.900
3/4	0.9666	0.4866	0.8939	0.8014	0.8052	1.022/0.900
4/4	0.5223	0.7562	0.7128	0.8581	0.9560	1.061/0.900

End-of-window fidelity to the Stage-1 latent starts high and erodes over the run: 0.9401 at window 1 falling to 0.8581 at window 4 (trend-summary WARN; min 0.8014 at window 3), and first-frame fidelity falls further still ($0.9834 \rightarrow 0.7128$, WARN) — refinement drifts furthest from the Stage-1 input late in the clip rather than converging back. Intra-window similarity dips mid-run (0.487 at window 3) against 0.76–0.82 elsewhere. As in T2V, the distilled refinement adds spatial high-frequency energy at every window (pre-anchor hf share 0.45–0.51 against Stage-1 references of 0.34–0.39) and the per-window detail anchor pulls it back with g_{hi} at the 0.90 floor cap throughout; because each window is referenced to its own Stage-1 slice, the trim cannot accumulate into progressive softening. Video seam continuity ranges 0.52–0.97, weakest at the window-3 $\rightarrow$ 4 seam (0.5223, $t = 40.0$ s). The audio timeline is bit-identical to Stage 1 (frozen_verify_sim= 1.0 per window); the Aud_bnd column re-measures the Stage-1 audio’s own chunk continuity where the video windows happen to cut (0.59–0.96 at $t = 13.33/26.67/40.00$ s). The assembled refined latents

are [1, 128, 160, 22, 40] (video, std 0.909) and [1, 8, 1333, 16] (audio, frozen), with no NaN/Inf.

6.4 Computational Overhead

Table 12: Wall-clock timing for the I2V run on a 16 GB GPU (RTX-class, CPU block-streaming); the T2V run’s per-pass timings match to within a few percent (its Stage 1 wall-clock is $\approx 2\times$ higher because `measure_g=on` runs a second, diagnostic denoising pass per chunk, discarded after measurement). Stage 1 runs three transformer passes per step (split CFG, STG); Stage 2 runs a single video refinement pass per window (audio frozen).

Stage / pass	Chunk	Steps	s/it	Time	Tokens
S1	1/10	20	20.6	≈ 6.9 min	—
S1	2–10/10	20	26.8–33.7	≈ 9 –11 min each	—
S2 video	1/4	3	76.9	≈ 3.8 min	35,200
S2 video	2–4/4	3	113.7–116.4	≈ 5.7 min each	46,640

Total generation time: Stage 1 ≈ 95 min + Stage 2 ≈ 21 min $\approx$ **116 min** for a ≈ 53 -second clip. The dominant cost driver is the three-pass guidance in Stage 1 — each denoising step runs three transformer passes (conditional, unconditional, STG-perturbed) rather than one. Chunk 1 is cheaper than the continuations (18.3 vs. 24.2–24.4 s/it) because it carries no 8-frame overlap context (16 vs. 24 latent frames in the window). In Stage 2 the per-window cost rises $\approx 50\%$ after window 1 (68.5 $\rightarrow$ 102–103 s/it) against a token-count rise of $\approx 32\%$: as the run proceeds an increasing share of the model is pushed to CPU (offloaded weights grow from ≈ 5.7 GB to ≈ 7.8 GB across the pass), so the slowdown is disproportionate to the token increase and is an offload-pressure effect rather than a compute-scaling one.

The T2V run costs the same as I2V to within rounding: per-chunk and per-window s/it match to a few percent at every position (e.g. Stage-1 chunk 1: 18.2 vs. 18.3 s/it; Stage-2 window 3: 105.1 vs. 102.8 s/it), confirming that the guide image changes neither the token count nor the compute cost — only the conditioning. The Stage-2 offload-pressure slowdown reproduces at the same windows in both runs, supporting that it is a property of the refinement pipeline rather than of run content.

7 Discussion

7.1 Effect of Chunk Length and Content Regime

Both runs used `new_lf = 16` (≈ 5.3 s per chunk), within the 21-frame safe limit established in earlier open-loop testing, streamed as ten chunks under the active guidance stack. In neither run does boundary similarity degrade with chunk index —

in both it *improves* first-to-last: I2V rises from an early floor of 0.835 (chunk 3) to 0.94–0.98 across the final five seams, and T2V from a 0.75–0.86 mid-run band to its best seam last (0.983 at chunk 10). The corrected loop does not compound error over this horizon in either conditioning mode; where continuity dips, it dips at isolated content events rather than accumulating — I2V’s floor sits early (chunk 3) and T2V’s mid-run (chunk 8), and both recover. The two runs exercise opposite content regimes — I2V a stable anchored scene (intra-chunk similarity plateauing at 0.91–0.94 in the final chunks), T2V sustained motion (intra swinging 0.13–0.88 with no plateau) — and the correction stack scales with the measured excess: gentle in both modes (I2V band gains ≥ 0.969 , detail-anchor gains within $+6.1\%/ -8.8\%$ of unity; T2V band-2 to 0.9716, g_{hi} bottoming at 0.919, never at the 0.90 floor).

7.2 Identity Tracking and Boundary Self-Consistency

Chunk 2’s `identity_sim` (0.9126) equals its `boundary_sim` exactly. This is not a coincidence but a *self-consistency check*: the anchor was seeded from chunk 1’s last frame, and the SLB passes that same frame as the first latent frame of chunk 2’s conditioning, so the anchor query and the boundary query compare the same source frame (chunk 1 frame 15) against the same target frame (chunk 2 frame 0). Both metrics agreeing validates that the anchor bank and the SLB operate on the same reference, and that the τ_c re-noising at the boundary introduces only minimal distributional shift. For later chunks the bank holds multiple entries and the identity query selects a different anchor than the boundary, so the two metrics diverge, as expected (e.g. I2V chunk 6: identity 0.940 vs boundary 0.978). The same self-consistency check holds in the unrelated T2V run at the same chunk (identity 0.8553 = boundary 0.8553, Table 3), confirming the mechanism is a property of the anchor/SLB bookkeeping rather than of any particular content.

Object-level identity. The metrics above are computed from mean-pooled channel features or low-band spatial maps and are therefore *object-blind*: a vehicle or person can be replaced by a different instance across a chunk boundary while boundary and identity cosines remain above 0.95. This failure class was observed perceptually and is invisible to the pooled telemetry. The `object_identity` lever (Section 5.4) closes the observation gap: it decodes the two seam frames at each chunk boundary and reports a spatially-resolved grid-cell cosine (`vid_obj`), so a regional content swap drops the minimum cell score even when the global mean stays high. The I2V run reported here was executed with the lever on: the grid-cell mean rises $0.952 \rightarrow 0.995$ across the nine seams (Table 8), and the per-seam minimum cell score stays within 0.84–0.98 — its lowest reading (0.836) sits at the weakest early seam region (chunks 2–3, where the pooled boundary metric also floors), so on this run the spatially-resolved metric corroborates the pooled one and flags no regional object swap. One run without a detected swap does not bound the failure rate; the mitigation for object-critical content remains textual — one prompt per chunk via **CLSSScenePrompts**, re-describing the persistent objects explicitly.

7.3 Audio Coherence and the Role of Visual Context

A natural hypothesis is that a guide image shapes streaming audio behaviour: in LTX-2.3’s joint audio-video transformer, video and audio tokens share self-attention layers, so a strong single-frame visual prior could constrain audio generation implicitly through cross-modal attention — for better (a coherence anchor) or worse (a repetition template). The matched pair does not support the hypothesis in either direction. Chunk-1 audio coherence starts *higher* without the guide image (T2V 0.800 $\rightarrow$ 0.669 vs. I2V 0.376 $\rightarrow$ 0.684 within chunk 1), and the envelope-repetition signature is absent in *both* modes: `env_corr` never crosses the 0.7 repetition flag anywhere (T2V peaks at 0.347; I2V trends 0.248 $\rightarrow$ 0.077, falling). What differences remain track the prompted *sound content* — T2V’s loud continuous ambience (RMS reference 0.996) naturally self-correlates within and across chunks, while I2V’s quieter, more varied scene audio (RMS reference 0.622) does not — rather than the presence or absence of a visual prior. The practical conclusion is negative but useful: visual conditioning is not a lever on streaming audio behaviour at this scale, and the audio failure modes of Section 7.6 must be addressed in the audio path itself.

7.4 Stability Margin

With $\rho_{\text{loop}} = 0.570$, the CLSS feedback loop provides a $\approx 43\%$ gain reduction per chunk before accounting for the model’s open-loop gain g . In the I2V run the spectral-shrinkage correction was light throughout — band-2/3 gains of 0.969–0.998, a $\approx 1\text{--}3\%$ high-frequency energy excess over the EMA reference. Taking the typical $\approx 1\text{--}3\%$ per-chunk growth as a rough estimate of the open-loop gain in the spectral channel gives $g \approx 1.02$ and $\rho_{\text{closed}} \approx 0.57 \times 1.02 \approx 0.58$, comfortably below the stability boundary of 1. The T2V run sits at the same operating point on this channel: band-2 gains span 0.972–0.996, again a $\approx 1\text{--}3\%$ typical excess, giving the same $g \approx 1.02$ and $\rho_{\text{closed}} \approx 0.58$ — the margin appears to be a property of the correction stack rather than of the conditioning mode. The raw audio, however, shows a spectral drift the correction stack does not reach: the Stage-1 trend summary’s `aud_hf` statistic (a ratio of high-band to reference energy, not covered by the video-side machinery of Sections 3.3 and 3.5) still grows in T2V — 0.607 $\rightarrow$ 0.835, inside the trend checker’s stability tolerance, with the top frequency bin individually reaching $1.34\times$ the chunk-1 reference by the final chunks — though it *falls* in I2V (0.974 $\rightarrow$ 0.897). This is audio-domain high-frequency energy growth with no video-channel analogue: the video hf channel is closed by the detail anchor, while no current correction targets the audio frequency distribution (the RMS anchor constrains only global loudness).

Measured g_{SLB} : the proxy was off by an order of magnitude. A same-seed companion I2V run executed with `measure_g=on` (under the previous audio SLB schedule; the perturbation protocol of Section 3.6 is video-side) gives the first live reading of Eq. (10): g_{SLB} falls from 39.6 (chunk 2) to 19.6 (chunk 10) with secondary peaks at chunks 5 (34.3) and 9 (36.7), a mean of ≈ 26.7 across the nine measured seams. The reported T2V run carries the same measurement in-run (Table 3): g_{SLB}

spans 15.8–87.0 (chunk-2 maximum 87.0, chunk-10 value 23.6, mean ≈ 34.6) — the same order of magnitude in the opposite conditioning mode. Two consequences follow. First, the informal band-energy proxy cited above ($g \approx 1.02$ – 1.05) is falsified as an estimate of transformer sensitivity: it measured the *residual* spectral drift surviving the correction stack, not the transformer’s raw response to a boundary perturbation — the two quantities differ by a factor of ≈ 25 . Second, the naive product $\rho_{\text{loop}} \times g_{\text{SLB}} \approx 0.57 \times 26.7 \approx 15$ sits an order of magnitude above the stability boundary of 1, yet that companion run was observably stable (video std $\Delta + 0.007$ over ten chunks, boundary similarity *rising* $0.909 \rightarrow 0.977$). The linear gain model of Section 3.6 therefore does not describe the closed loop asymptotically: the one-step sensitivity g_{SLB} includes legitimate content re-interpretation of a 1% boundary perturbation, which the next denoise pulls back toward the data manifold, and that contraction — absent from the model — is what actually bounds drift. The model remains useful as a local design guide (which correction attenuates which channel, and by how much), but the stability margin is an empirical property of the loop, not a computable product of per-channel gains. Its simplifying assumptions (constant g , independent multiplicative attenuators) likewise abstract away the higher-order interactions between corrections, guidance, and scene content; a Lyapunov-style treatment or a Monte Carlo sweep over perturbation seeds and chunk positions would be needed for a rigorous account.

7.5 Stage-2 Identity Metric Ambiguity

Stage 2 lacks an anchor bank; its `identity_sim` reduces to comparing each window against the fixed first-window reference. On a single moving scene this conflates ordinary content motion with drift: in I2V, where the reference is the guide-image region, windows 2–4 read 0.59, 0.56, 0.29 — a decaying plateau reflecting that the video has moved away from its opening frames, not instability. In T2V, where window 1 has no anchor to begin with, the same comparison instead wanders just above zero (0.20, 0.03, 0.28 across windows 2–4) — consistent with window 1 itself being only weakly self-correlated (Table 6, intra 0.21), so the fixed reference is diffuse and later windows have nothing sharp to be “close to” or “far from” in the first place. Adding a scene-aware anchor bank to Stage 2 would make this metric interpretable in either conditioning mode; as reported it should be read only as “distance from the opening,” not as a drift signal.

7.6 Measured Failure Modes at Longer Horizons

Per-frame telemetry across extended runs — including the ten-chunk, ≈ 53 -second T2V and I2V runs reported above and earlier 13-lf and 31-lf runs — exposes two repetition mechanisms with distinct signatures, plus the Stage-2 audio design consequence.

Loudness-envelope repetition: carrier found, lock broken. Earlier pipeline versions repeated a per-chunk loudness gesture in every configuration tested: `env_corr` locked above the 0.7 repetition flag (sustained 0.76–0.96 in the previous T2V pipeline;

0.87–0.97 in the previous I2V pipeline), audible as the same swell replaying every chunk. A same-seed I2V A/B isolated the carrier: the audio SLB passes the previous chunk’s tail nearly verbatim, and the gesture circulates through it — disabling the SLB dropped `env_corr` to 0.45–0.51 but collapsed audio boundary similarity to 0.28–0.58 with high-frequency runaway ($0.54 \rightarrow 1.25$), audible as sudden jumps and degraded speech. The final pipeline therefore keeps the SLB but attacks the carry: the audio overlap is re-noised on a hotter schedule ($\tau_c \approx 0.15 \rightarrow 0.28$ over the run, versus $\approx 0.05 \rightarrow 0.12$ previously) and the audio context is envelope-flattened before conditioning (Section 5.3). In the reported I2V run the lock is gone: `env_corr` stays below the flag at every seam (trend $0.248 \rightarrow 0.077$, *falling*), the loudness peak no longer phase-locks, and the repetition is not audible — while continuity is preserved (SLB-honored ≥ 0.9998 despite the re-noise; audio boundary 0.76–0.94; RMS-anchor corrections fall back to trims of at most 7.4%). The reported T2V run is likewise clean: `env_corr` never exceeds 0.347 at any seam and its loudness peaks scatter (mode covers 2/10 chunks). The earlier intervention record is consistent with this account: context flattening *alone* (previous schedule) left the lock at 0.97, so the re-noise schedule is the active ingredient; guidance and reference-perturbation variants (modality guidance raised, noise blending, length decay) changed nothing and added hiss. A prompt-driven baseline remains — every chunk denoises against the same static text, and even with no SLB the envelope correlation sat at ≈ 0.5 — but it stays below the audible threshold in the reported run.

Chunk-relative phase lock. An earlier symptom of the same mechanism was a hard *phase* lock: the audio energy peak at 94% of the window for eight consecutive 13-lf chunks. A per-chunk overlap phase-jitter lever was built against it and did scatter the peak, but it fought the SLB’s structural assumptions and was removed; under the previous audio schedule the lock re-emerged late (peaks pinning to one chunk-relative frame from chunk 6 onward), but under the raised re-noise schedule it does not form: the reported I2V run’s peaks scatter across frames and modes (38, 27, 84, 18, 81, 78, 129, 129, 130, 78; mode covers 3/10 chunks within ± 1). Phase lock and envelope correlation were two readings of the same gesture replaying, and both are now quiet.

A ~ 4 -second video layout oscillation. Coarse scene layout departs from the scene reference and partially returns with a ≈ 4 s wall-clock period that is *invariant to chunk length*: at 13-lf (4.3 s) chunks the dip lands at new-frame 5–7 in every chunk (appearing seam-locked), while at 31-lf (9.9 s) chunks it recurs within the chunk at the same seconds-period. Context strength and context length (overlap 3/5/8) were tested and neither removes it; the oscillation appears intrinsic to the model’s trained temporal horizon. In the reported 16-lf runs the layout-argmin detector reads unlocked in both modes (I2V mode 3 covering 4/10 chunks within ± 1 , T2V mode 14 covering 4/10) — at overlap 8 the strict per-chunk swing-and-return does not phase-lock to the seam grid, though global layout still wanders (largest per-frame layout drops of -0.64 in I2V and -0.35 in T2V, spread across the first half of each run rather

than concentrated at one seam). The natural corrective — conditioning generation on the anchor bank, which already tracks long-range identity — has no known safe wiring: the reference library’s append-style `VideoConditionByKeyframeIndex` reproduces the AV audio corruption diagnosed for the i2v guide path (Section 5.5), and a low-strength in-place replacement at the first new frame produced visible content *morphing*: anchor retrieval returns the most-similar *non-redundant* anchor, and the redundancy filter excludes precisely those anchors that match the current overlap, so the retrieved frame is guaranteed to differ from the content being continued — the injection drags the chunk back toward earlier scene content. Conditioning the joint audio-video model on a temporally-displaced keyframe without either corrupting audio (append) or morphing video (in-place replace) is an unsolved design problem; the oscillation above remains unaddressed.

Stage-2 audio is frozen by design. The freeze (Section 5.2) is the outcome of a measured escalation. With per-window audio re-rendering at high denoise (σ -scale 1.0), $\approx 45\%$ of the audio is re-rolled per window and the re-rolls disagree at every window boundary (boundary cosine 0.32–0.42) *regardless* of the audio SLB length — the SLB pins the seam frame, not the style of the re-roll. At denoise 0.4 the pass polishes instead (seed similarity 0.79–0.81) yet seams still disagreed at 0.59–0.72: polishing narrows the disagreement but cannot close it while windows are denoised independently. Freezing (mask 0) closes it exactly — Stage 2 introduces no audio seams at all, and the only audio discontinuities in the final output are Stage 1’s own chunk boundaries. Rule: audio is either generated in one window or not re-rolled at all; any multi-window audio denoise re-seams the timeline it was meant to improve.

Audio high-frequency growth. The trend-summary’s `aud_hf` statistic (high-band audio energy relative to the chunk-1 reference) still grows in the reported T2V run: $0.607 \rightarrow 0.835$ — inside the trend checker’s stability tolerance, with the top frequency bin individually reaching $1.34\times$ reference by the final chunks. The reported I2V run shows *no* growth ($0.974 \rightarrow 0.897$, falling); whether the raised audio re-noise schedule or the flattened context suppresses the growth is not isolated, and the channel still has no dedicated correction: the video-side spectral shrinkage (Section 3.3) and detail anchor (Section 3.5) operate on the video latent only, and the audio-side RMS anchor (Section 5.3) constrains global loudness but not the frequency distribution. An audio-side two-band anchor, mirroring Section 3.5 but applied to the audio latent’s frequency axis, is the natural next mechanism to try; an earlier attempt at audio high-band shrinkage was reverted for audible artefacts, so any such correction must be validated by ear, not by the statistic it improves.

7.7 Scope and Generality

Single configuration, single seed, single hardware. The reported T2V and I2V results share one set of hyperparameters (`new_lf` = 16, `overlap` = 8, 10 chunks, 1 scene), one seed, and one hardware configuration (16 GB VRAM, RTX-class); they

differ only in the first-frame conditioning, by design, to isolate that variable. Every single-run comparison in this section (e.g. the phase-lock/guide-image association) is therefore a $n = 1$ -per-arm observation, not a statistically powered result. The ablation arms and the open-loop baseline are fully specified and exposed as node levers (Section 6, *Ablation and measurement levers*); ablation and seed-sweep runs are planned follow-up work.

Stability metrics vs. perceptual quality. The reported metrics—boundary cosine similarity, intra-chunk cosine similarity, audio within-chunk segment similarity—measure the *stability* of the latent representation across chunks. They do not assess the perceptual quality of the generated content, prompt adherence, or realism. High boundary similarity is a necessary but not sufficient condition for high video quality: standard quality metrics (FID, IS, CLIP similarity) and, ultimately, human evaluation would be required to characterise the generated video as a content generation system.

Hardware specificity. The GGUF 4-bit quantisation and CPU block-streaming configuration was designed for a specific VRAM budget (16 GB). Performance, latency, and numerical behaviour may differ on GPUs with more VRAM (where block-streaming is unnecessary) or with different quantisation schemes (e.g., Q8 or FP8).

8 Related Work

Long video generation. StreamingT2V [3] and FIFO-Diffusion [4] generate long videos by sliding-window denoising. Gen-L-Video [5] uses temporal attention patching. FreeNoise [1] reschedules the initial noise (window-based fusion of a shuffled, reused noise sequence) to extend a pretrained model’s temporal attention window without retraining or sliding-window denoising at all — a tuning-free approach to the same underlying problem (fixed training-time context length) that CLSS addresses by chunking with closed-loop correction instead. VidToMe [2] merges self-attention tokens across frames to improve temporal consistency in zero-shot video editing, operating within a single generation pass rather than across autoregressively-generated chunks. CLSS differs from all of these by applying closed-loop corrections specifically designed for the latent-streaming regime of large audio-video models, where each chunk is a genuinely separate denoising process conditioned on the previous chunk’s output — the exposure-bias setting FreeNoise and VidToMe do not target.

Exposure bias in diffusion. The exposure bias problem in autoregressive models is well-studied [6]. For diffusion models, Ning et al. [7] and Salimans & Ho [8] address related distributional mismatch issues during training, but inference-time corrections for streaming have received less attention.

Statistical normalisation for consistency. Adaptive Instance Normalisation (AdaIN) [9] was introduced for style transfer, where the goal is to match the mean and standard

deviation of a content feature map to those of a style reference. EMA-based statistical normalisation across video frames has also appeared in temporal consistency methods. In CLSS, AdaIN is repurposed as a *drift correction* applied to denoised latents between chunks, using the cross-chunk EMA as the style reference. The combination of EMA tracking with AdaIN blending (rather than full replacement) is motivated by the streaming setting, where the reference must itself evolve to allow intentional scene changes.

Frequency-domain video editing. Yang et al. [10] and Ceylan et al. [11] use frequency-domain constraints for temporal consistency in video editing. CLSS applies frequency shrinkage in the *latent* domain to a streaming generation setting, targeting energy accumulation rather than pixel-domain flicker.

9 Conclusion

We presented CLSS, a five-component closed-loop streaming synthesis framework that enables stable long audio-video generation with LTX-2.3 22B on consumer 16 GB VRAM hardware. The corrections — calibrated context re-noising, EMA-tracked AdaIN, frequency-band soft shrinkage, a dynamic anchor bank, and a two-band spatial detail anchor — achieve $\rho_{\text{loop}} = 0.570$; driven by a reference-parity multi-modal guider (split per-modality CFG and STG) that correctly handles ComfyUI’s packed audio-video latent representation, the pipeline held Stage 1 video boundary similarity at 0.84–0.98 (I2V) and 0.75–0.98 (T2V) across ten chunks of matched ≈ 53 -second single-scene clips, held the high-frequency energy share flat end-to-end in both conditioning modes, and held audio loudness to its chunk-1 reference (output RMS within $\pm 5\%$, absorbing raw per-chunk dips to -18% in T2V). A two-stage pipeline (streaming Stage 1 + windowed distilled Stage 2 with per-window detail anchoring and frozen audio) delivers a fully refined $2\times$ -resolution audio-video output in ≈ 116 minutes per clip on a single 16 GB GPU, essentially independent of conditioning mode.

Running matched T2V and I2V generations under an identical seed and guidance stack, differing only in the presence of a guide image, shows that visual conditioning is not a lever on streaming audio behaviour: chunk-1 audio coherence, envelope repetition, and loudness stability all track the prompted sound content rather than the guide image, while the guide image’s clear effects are on the video side (a stable anchored scene with an early boundary floor of 0.835 rising to 0.94–0.98, versus T2V’s looser 0.75–0.86 mid-run band closing at 0.983 under sustained motion). Two repetition failure modes dominated earlier pipeline versions; both are now characterised, and one is resolved. The per-chunk audio loudness gesture — `env_corr` locked above 0.7 in every earlier configuration, audible as the same swell replaying every chunk — is broken by re-noising the audio overlap on a hotter schedule with an envelope-flattened context (Section 7.6): `env_corr` stays below 0.35 at every seam in both reported runs while seam continuity is preserved. The chunk-length-invariant ≈ 4 -second video layout oscillation persists but no longer phase-locks to the seam grid at overlap 8; it appears intrinsic to the model’s trained temporal window. All of

this is visible in the pipeline’s per-chunk telemetry — unconditional settings dumps, drift-event tables, and a phase-lock detector — which is designed to localise such effects in wall-clock time and latent-frame position, and to make a silently inactive guidance path impossible to miss (a guidance term that does not change per-step wall-clock time is not running).

Object-level identity persistence across chunk boundaries, invisible to the current structural metrics in both conditioning modes, remains the principal open problem for content-level quality.

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